

Deep Generative Models

Lecture 14

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AI Masters

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Recap of Previous Lecture

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

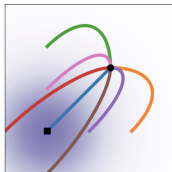
$$p_t(\mathbf{x}|\mathbf{x}_1) = q_{1-t}(\mathbf{x}|\mathbf{x}_0 = \mathbf{x}_1)$$

Variance Exploding SDE

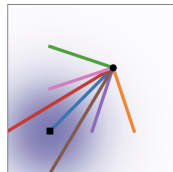
$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(\mathbf{x}_1, \sigma_{1-t}^2 \mathbf{I}) \Rightarrow \mathbf{v}(\mathbf{x}, \mathbf{x}_1, t) = -\frac{\sigma'_{1-t}}{\sigma_{1-t}}(\mathbf{x}_t - \mathbf{x}_1)$$

Variance Preserving SDE

$$p_t(\mathbf{x}_t|\mathbf{x}_1) = \mathcal{N}(\alpha_{1-t}\mathbf{x}_1, (1 - \alpha_{1-t}^2)\mathbf{I}) \Rightarrow \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\alpha'_{1-t}}{1 - \alpha_{1-t}^2} \cdot (\alpha_{1-t}\mathbf{x}_t - \mathbf{x}_1)$$



Diffusion



OT

Recap of Previous Lecture

Continuous state space

- ▶ **Discrete time** $t \in \{0, 1, \dots, T\} \Rightarrow$ **DDPM / NCSN.**
- ▶ **Continuous time** $t \in [0, 1] \Rightarrow$ **Score-based SDE models.**

Discrete state space

- ▶ **Discrete time** $t \in \{0, 1, \dots, T\}.$
- ▶ **Continuous time** $t \in [0, 1].$

Key advantages of discrete diffusion

- ▶ Parallel generation
- ▶ Flexible infilling
- ▶ Robustness
- ▶ Unified framework

Recap of Previous Lecture

Discrete Diffusion Markov Chain

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \text{Cat}(\mathbf{Q}_t \mathbf{x}_{t-1}),$$

Each $\mathbf{x}_t \in \{0, 1\}^K$ is a **one-hot vector** encoding the categorical state (it is just one token).

Transition Matrix

$$[\mathbf{Q}_t]_{ij} = q(x_t = i | x_{t-1} = j), \quad \sum_{i=1}^K [\mathbf{Q}_t]_{ij} = 1.$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \text{Cat}(\mathbf{Q}_{1:t} \mathbf{x}_0), \quad \mathbf{Q}_{1:t} = \mathbf{Q}_t \mathbf{Q}_{t-1} \cdots \mathbf{Q}_1.$$

- ▶ The choice of $\mathbf{Q}_t$ determines how information is erased and what the stationary distribution becomes.
- ▶ $\mathbf{Q}_t$ and $\mathbf{Q}_{1:t}$ should be easy to compute for each t .

Recap of Previous Lecture

Uniform vs. Absorbing Transition Matrix

Aspect	Uniform Diffusion	Absorbing Diffusion
$\mathbf{Q}_t$	$(1 - \beta_t)\mathbf{I} + \beta_t\mathbf{U}$	$(1 - \beta_t)\mathbf{I} + \beta_t\mathbf{e}_m\mathbf{1}^\top$
$\mathbf{Q}_{1:t}$	$\bar{\alpha}_t\mathbf{I} + (1 - \bar{\alpha}_t)\mathbf{U}$	$\bar{\alpha}_t\mathbf{I} + (1 - \bar{\alpha}_t)\mathbf{e}_m\mathbf{1}^\top$
$\mathbf{Q}_{1:\infty}$	$\mathbf{U}$	$\text{Cat}(\mathbf{e}_m)$
Interpretation	Random replacement	Gradual masking of tokens
Application	Image diffusion	Text diffusion $\approx$ Masked LM

Observation

Both schemes gradually destroy information, but differ in their stationary limit. Absorbing diffusion bridges diffusion and masked-language-model objectives.

Recap of Previous Lecture

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}$$

Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^{\top} \mathbf{Q}_{1:t} \mathbf{x}_0}\right).$$

- ▶ Both $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and $q(\mathbf{x}_t|\mathbf{x}_0)$ are known analytically from the forward process.
- ▶ The reverse process $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is a learned categorical distribution:

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}(\boldsymbol{\pi}_{\theta}(\mathbf{x}_t, t)).$$

Recap of Previous Lecture

ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)).$$

Categorical KL

$$D_{\text{KL}}(\text{Cat}(\mathbf{q}) \parallel \text{Cat}(\mathbf{p})) = \sum_{k=1}^K q_k \log \frac{q_k}{p_k} = H(\mathbf{q}, \mathbf{p}) - H(\mathbf{q}),$$

- ▶ $H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0))$ is a constant w.r.t. θ .
- ▶ $H(\mathbf{q}, \mathbf{p}) = -\sum_k q_k \log p_k$ is a **cross-entropy loss**.

Therefore, minimizing $\mathcal{L}_t$ w.r.t. θ is equivalent to minimizing

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} H(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0), p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)).$$

Outline

1. Discrete Diffusion

From Token to Sequence

Absorbing Diffusion

Continuous-Time Formulation

2. Course Overview

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From Token to Sequence

One-hot sequence representation

$$\mathbf{x}_t \in \{0, 1\}^K \quad \Leftrightarrow \quad \mathbf{X}_t \in \{0, 1\}^{K \times m}$$

Here $\mathbf{X}_t$ is a one-hot representation of a sequence of tokens.

From Token to Sequence

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Independent Token-wise Forward Process

$$q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \prod_{i=1}^m q(\mathbf{x}_t^i | \mathbf{x}_{t-1}^i) = \text{Cat}(\mathbf{Q}_t \mathbf{X}_{t-1})$$

- ▶ Each position i evolves according to its own Markov chain.
- ▶ Often the same transition matrix $\mathbf{Q}_t$ is shared across i .

From Token to Sequence

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Continuous Diffusion Analogy

- ▶ In Gaussian DDPMs with diagonal covariance, noise is independent per pixel.
- ▶ Structure is not in the noise; it is learned by the reverse model.

From Token to Sequence

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Conditioned Reverse Distribution

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \text{Cat} \left(\frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1} \mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t} \mathbf{x}_0} \right).$$

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- ▶ All distributions defined by the forward process are factorized.
- ▶ Dependence appears in the learned reverse model.

Reverse Model for Sequence

$$p_{\theta}(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^m p_{\theta}(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

- ▶ The output factorizes (parallel prediction across positions).
- ▶ Each factor conditions on the entire noisy sequence $\mathbf{X}_t$.
- ▶ This is exactly the **masked language modeling** pattern.

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Objective: $\mathcal{L}_t$ term

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Final objective: masked LM

$$\mathcal{L} = \sum_{t=1}^T \sum_{i=1}^m \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[-\log p_{\theta}(\mathbf{x}_0^i|\mathbf{X}_t) \right].$$

Outline

1. Discrete Diffusion

From Token to Sequence

Absorbing Diffusion

Continuous-Time Formulation

2. Course Overview

Absorbing Diffusion: Forward Process

Let's restrict to the case of absorbing transition matrix.

$$\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t \mathbf{e}_m \mathbf{1}^\top, \quad \bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s).$$

$$\mathbf{Q}_{1:t} = \bar{\alpha}_t \mathbf{I} + (1 - \bar{\alpha}_t) \mathbf{e}_m \mathbf{1}^\top.$$

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$$\mathbf{Q}_t = \begin{pmatrix} 1 - \beta_t & 0 & 0 \\ 0 & 1 - \beta_t & 0 \\ \beta_t & \beta_t & 1 \end{pmatrix} \Rightarrow \text{the masked state is absorbing.}$$

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What happens in the conditioned reverse process $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$?

Absorbing Diffusion

Conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} [\mathbf{x}_{t-1} = \mathbf{x}_t], & \text{if } \mathbf{x}_t \neq \mathbf{e}_m, \\ \rho_t[\mathbf{x}_{t-1} = \mathbf{x}_0] + (1 - \rho_t)[\mathbf{x}_{t-1} = \mathbf{e}_m], & \text{if } \mathbf{x}_t = \mathbf{e}_m, \end{cases}$$

where

$$\rho_t = \frac{\beta_t \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}.$$

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- ▶ If $\mathbf{x}_t \neq \mathbf{e}_m$, then the token must be unchanged: $\mathbf{x}_{t-1} = \mathbf{x}_t$.
- ▶ Observing an unmasked token at time t fixes the entire history: $\mathbf{x}_{t-1} = \mathbf{x}_t = \cdots = \mathbf{x}_0$.

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- ▶ Observing an unmasked token at time t fixes the entire history: $\mathbf{x}_{t-1} = \mathbf{x}_t = \dots = \mathbf{x}_0$.
- ▶ If $\mathbf{x}_t = \mathbf{e}_m$, the previous token may be either clean or masked.
- ▶ With probability ρ_t , masking occurred exactly at step t (so $\mathbf{x}_{t-1} = \mathbf{x}_0$).
- ▶ With probability $(1 - \rho_t)$, the token was already masked earlier (so $\mathbf{x}_{t-1} = \mathbf{e}_m$).

Absorbing Diffusion

Sequence Distribution

$$q(\mathbf{X}_{t-1}|\mathbf{X}_t, \mathbf{X}_0) = \prod_{i=1}^m q(\mathbf{x}_{t-1}^i|\mathbf{x}_t^i, \mathbf{x}_0^i).$$

Each position i has two possible cases:

$$\mathbf{x}_t^i \neq \mathbf{e}_m \Rightarrow \mathbf{x}_{t-1}^i = \mathbf{x}_t^i, \quad \mathbf{x}_t^i = \mathbf{e}_m \Rightarrow \mathbf{x}_{t-1}^i \in \{\mathbf{x}_0^i, \mathbf{e}_m\}.$$

Absorbing Diffusion

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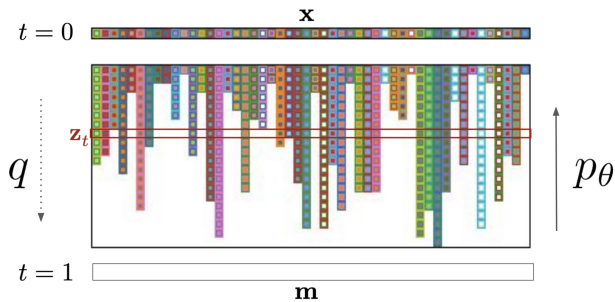
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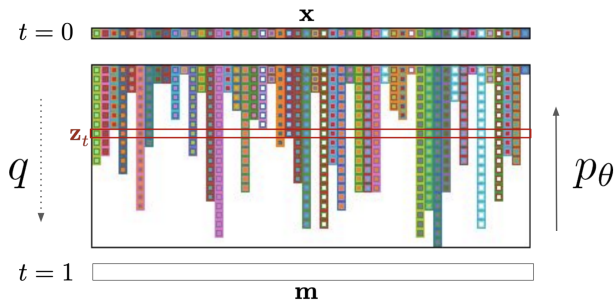
Interpretation

- ▶ The forward process produces **random partial observations** of the clean sequence.
- ▶ If a token is visible at time t , the reverse distribution is deterministic.
- ▶ Unmasked tokens yield a deterministic posterior and therefore contribute only a constant to the ELBO. Therefore, only masked tokens contribute to the training loss.

Absorbing Diffusion

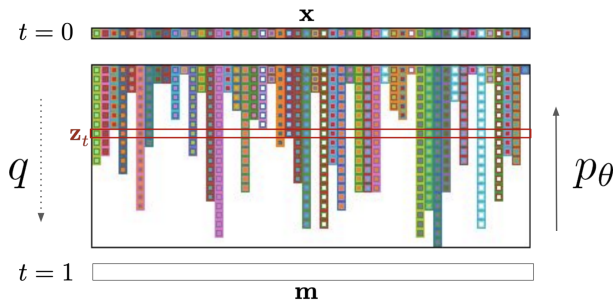


Absorbing Diffusion



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Absorbing Diffusion



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Objective: sequence-level $\mathcal{L}_t$

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \sum_{i=1}^m \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_\theta(\mathbf{x}_0^i|\mathbf{X}_t) \right] + \text{const}$$

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From Discrete Time to Mask Rate

In absorbing diffusion, the forward process is

$$q(\mathbf{x}_t|\mathbf{x}_0) = \bar{\alpha}_t[\mathbf{x}_t = \mathbf{x}_0] + (1 - \bar{\alpha}_t)[\mathbf{x}_t = \mathbf{e}_m].$$

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- The distribution depends on t only through the scalar

$$\lambda_t = 1 - \bar{\alpha}_t \in [0, 1].$$

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$$t \quad \Rightarrow \quad \lambda \in [0, 1].$$

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- ▶ We can therefore reparameterize the corruption level by

$$t \quad \Rightarrow \quad \lambda \in [0, 1].$$

- ▶ We directly define a family of corrupted distributions indexed by a continuous mask rate λ :

$$q(\mathbf{x}_\lambda|\mathbf{x}_0) = (1 - \lambda)[\mathbf{x}_\lambda = \mathbf{x}_0] + \lambda[\mathbf{x}_\lambda = \mathbf{e}_m].$$

Discrete ELBO Revisited

Recall the per-step ELBO term for absorbing diffusion:

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \sum_{i=1}^m \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_{\theta}(\mathbf{x}_0^i|\mathbf{X}_t) \right] + \text{const.}$$

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Replacing the discrete index t with the continuous mask rate λ , the training objective becomes

$$\mathcal{L} = \int_0^1 w(\lambda) \mathbb{E}_{q_\lambda(\mathbf{x}_\lambda|\mathbf{x}_0)} \sum_{i=1}^m [\mathbf{x}_\lambda^i = \mathbf{e}_m] [-\log p_\theta(\mathbf{x}_0^i|\mathbf{X}_\lambda)] d\lambda.$$

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Interpretation

Training corresponds to optimizing a **continuous mixture of masked language modeling objectives** with different mask rates.

Algorithm: Masked Diffusion Language Model (MDLM)

Training

1. Sample $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X})$ and $\lambda \sim w(\lambda)$.
2. Corrupt the sequence by independent masking:

$$\mathbf{x}_{\lambda}^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } \lambda, \\ \mathbf{x}_0^i, & \text{with prob. } 1 - \lambda, \end{cases} \quad i = 1, \dots, m.$$

3. Predict token distributions in parallel:

$$p_{\theta}(\mathbf{X}_0 | \mathbf{X}_{\lambda}) = \prod_{i=1}^m p_{\theta}(\mathbf{x}_0^i | \mathbf{X}_{\lambda}).$$

4. Compute the masked-CE loss:

$$\mathcal{L}(\theta) = \sum_{i=1}^m [\mathbf{x}_{\lambda}^i = \mathbf{e}_m] [-\log p_{\theta}(\mathbf{x}_0^i | \mathbf{X}_{\lambda})].$$

Algorithm: Masked Diffusion Language Model (MDLM)

Sampling

1. Initialize $\mathbf{X} \leftarrow \mathbf{e}_m \mathbf{1}^\top$ (fully masked).
2. For a decreasing schedule $\lambda_1 > \lambda_2 > \dots > \lambda_L$:
 - 2.1 Predict $p_\theta(\mathbf{x}^i | \mathbf{X})$ for all positions.
 - 2.2 Unmask a subset of positions to reach the next mask rate $\lambda_{\ell+1}$ (e.g., sample tokens for newly-unmasked positions).
3. Return the final sequence $\mathbf{X}$ (fully unmasked).

Outline

1. Discrete Diffusion

From Token to Sequence

Absorbing Diffusion

Continuous-Time Formulation

2. Course Overview

Course Overview: Problem Statement

Goal

Learn a generative model $p_{\theta}(\mathbf{x})$ that matches the data distribution $p_{\text{data}}(\mathbf{x})$.

Three similar lenses

- **Divergence minimization:**

$$\min_{\theta} D(p_{\text{data}} || p_{\theta}) \quad (\text{KL, JS, Wasserstein, etc.})$$

- **Likelihood-based modeling:** maximize $\log p_{\theta}(\mathbf{x})$ (NF, VAE, diffusion-as-VAE).
- **Score-based modeling:** learn $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ (DDPM / NCSN / SDE).

What We Covered: Part 1

Likelihood-based

► Autoregressive (L1):

$$p_{\theta}(\mathbf{x}) = \prod_{i=1}^n p_{\theta}(x_i | \mathbf{x}_{1:i})$$

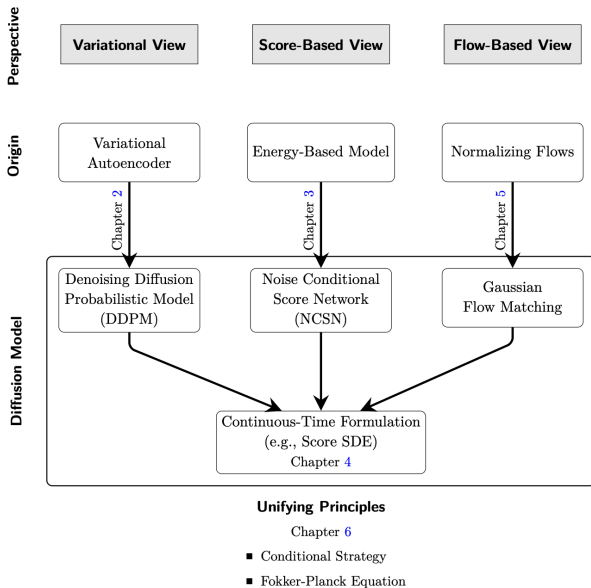
► Normalizing Flows (L2, L10):

$$\mathbf{x} = \mathbf{f}_{\theta}(\mathbf{z}), \quad \log p_{\theta}(\mathbf{x}) = \log p(\mathbf{z}) + \log \left| \det \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right|$$

► VAE (L3–L4):

$$\log p_{\theta}(\mathbf{x}) \geq \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})}[\log p_{\theta}(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) || p(\mathbf{z}))$$

Generative Models Timeline

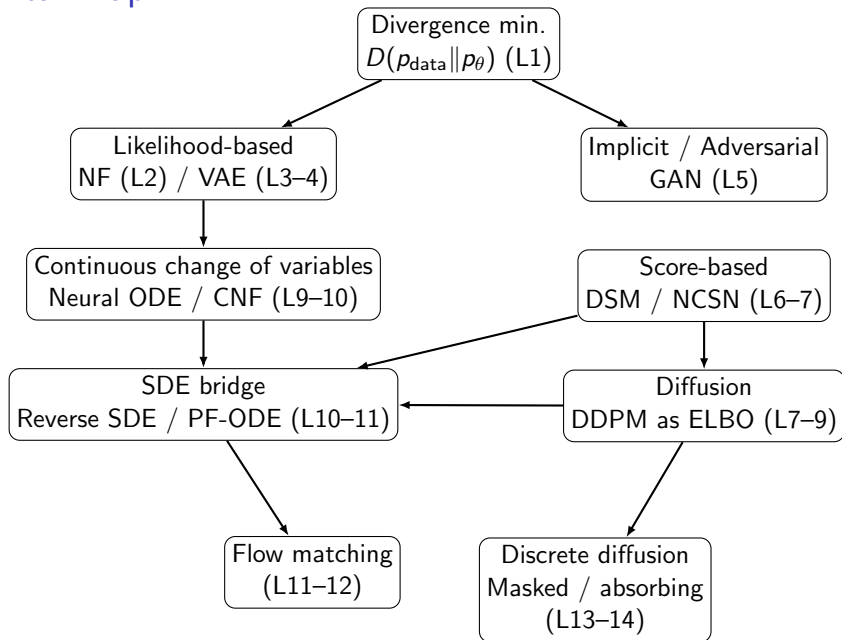


What We Covered: Part 2

Implicit / Score-based

- ▶ **GAN (L5)**: implicit p_θ , adversarial learning (close to JSD)
- ▶ **Score matching (L6)**: learn $\mathbf{s}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$
- ▶ **Diffusion / DDPM (L7–L9)**: forward noising $q(\mathbf{x}_t | \mathbf{x}_{t-1})$ + reverse denoising
- ▶ **SDE / Flow matching (L10–L12)**: reverse SDE $\leftrightarrow$ probability flow ODE, vector field (Neural ODE) / flow matching
- ▶ **Discrete diffusion (L13–L14)**: Markov chain on categorical states

Mental Map



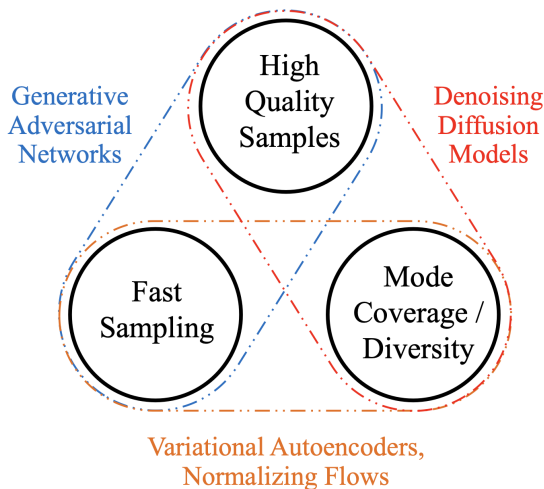
Comparison Cheat-Sheet: Part 1

Family	Likelihood	Training
AR	✓	stable CE
NF	✓	tricky architecture
VAE	lower bound	stable ELBO
GAN	×	unstable/minimax
Diffusion	bound / est.	stable
FM / ODE	est. / bound	stable
Discr. diff.	bound / CE-like	stable

Comparison Cheat-Sheet: Part 2

Family	Sampling	Best for
AR	slow (sequential)	text / discrete
NF	fast (1 step)	exact density, OOD
VAE	fast (1 step)	latent modelling
GAN	fast (1 step)	sharp images
Diffusion	slow (many steps)	high fidelity + diversity
FM / ODE	medium-fast	fewer steps + theory
Discr. diff.	iterative	sequences + bidirectional gen

Generative Learning Trilemma



Xiao Z., Kreis K., Vahdat A. Tackling the generative learning trilemma with denoising diffusion GANs, 2021

Generative Learning Trilemma

Rule of Thumb

- ▶ **Likelihood & Coverage $\Rightarrow$ AR / NF** exact density, no mode dropping, *slow sampling*
- ▶ **Likelihood & Fast Sampling $\Rightarrow$ VAE** tractable bound, fast, *blurry samples*
- ▶ **Sample Quality & Fast Sampling $\Rightarrow$ GAN** sharp samples, *no likelihood, mode collapse*
- ▶ **Quality & Coverage $\Rightarrow$ Diffusion** stable training, high fidelity, *slow sampling*
- ▶ **Quality & Faster Sampling $\Rightarrow$ FM / ODE** fewer steps, continuous flows, *approx. likelihood*
- ▶ **Discrete Structure & Coverage $\Rightarrow$ Discrete Diffusion** stable CE training, parallel denoising, *iterative decoding*

Summary

- ▶ For sequences, the forward process of discrete diffusion factorize over positions, but reverse process (the model p_{θ}) conditions on the whole context.
- ▶ In the absorbing case, tokens are either unchanged or masked; so only masked tokens contribute to the ELBO loss.
- ▶ The discrete ELBO reduces to a MLM objective.
- ▶ Reparameterizing time by the mask rate $\lambda \in [0, 1]$ yields a continuous mixture of MLM losses.
- ▶ MDLM sampling performs iterative parallel refinement from fully masked to fully unmasked sequences.
- ▶ No generative model is strictly better than all others: different methods occupy different corners of the generative learning trilemma and come with unavoidable disadvantages.